DATABASE JOINS

OVERVIEW

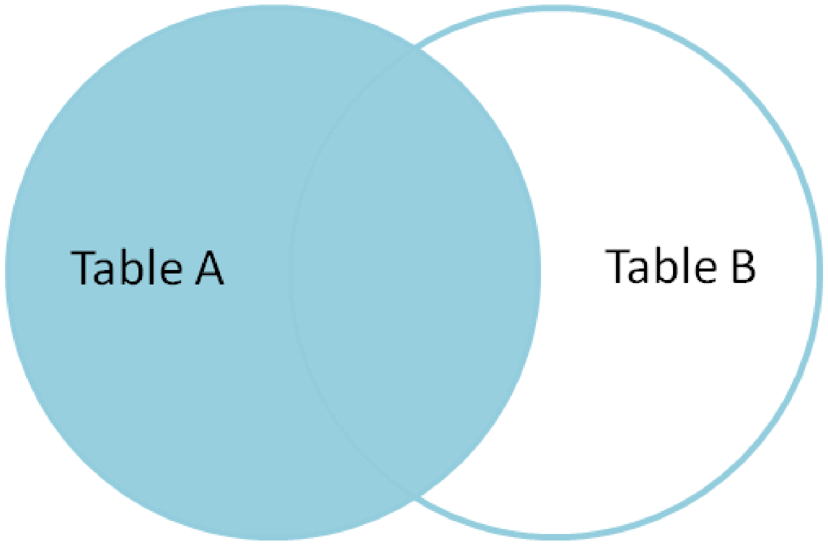
- What are Database Joins?
- Left Join
- Right Join
- Inner Join
- Full Outer Join

WHAT ARE DATABASE JOINS?

- A Join clause is used to combine rows from two or more tables, based on a related column between them
- Joins are often used by database professionals to improve and ensure data integrity
- We have four different types of joins in SQL:
  - Left Join
  - Right Join
  - Inner Join
  - Outer Join

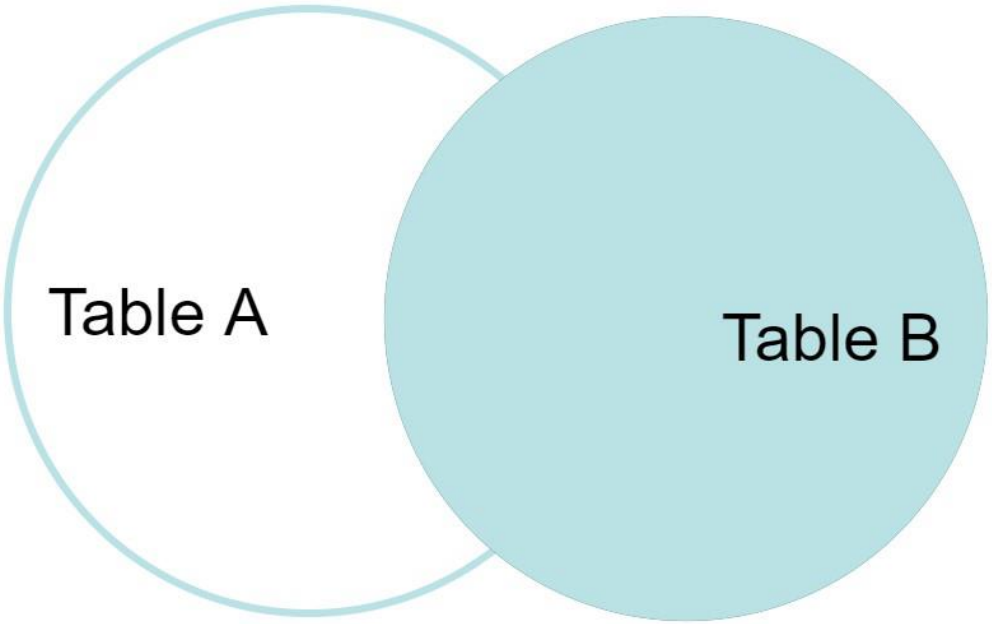
LEFT JOIN

- The left join returns all records from the left (1st) table, and the matching records from the right (2nd) table
- For the rows for which there is no matching row on the right side, the result-set will contain *null*
- Syntax: SELECT table1.column1, table1.column2, table2.column1... FROM table 1 LEFT JOIN table2 ON table1.matching_column = table2.matching_column;



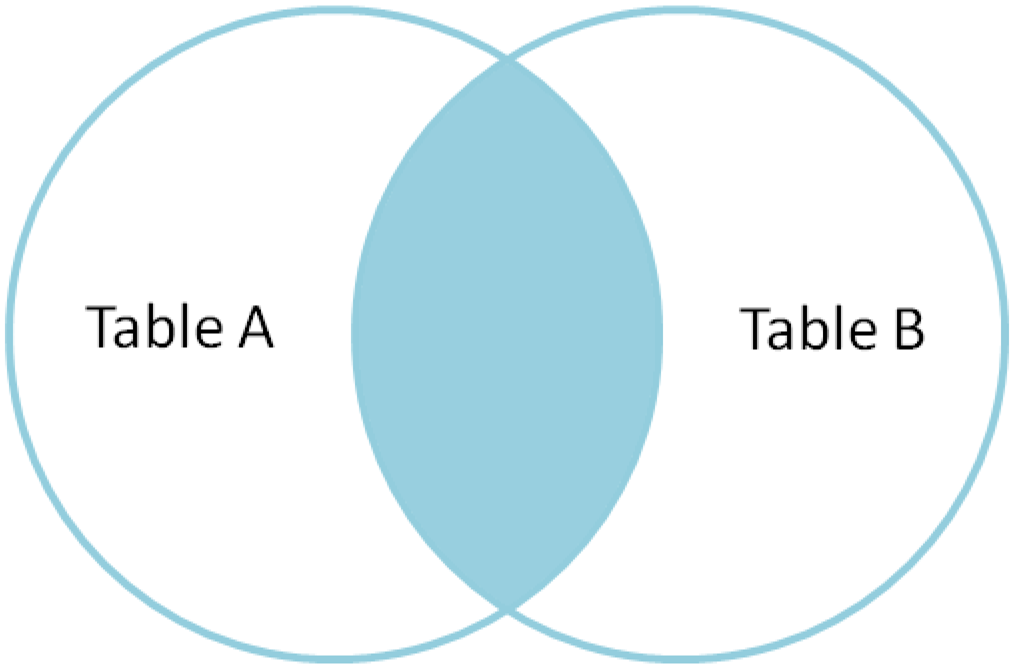
RIGHT JOIN

- The right join returns all records from the right (2nd) table, and the matching records from the left (1st) table
- For rows for which there is no matching row on the left side, the result-set will contain null
- Syntax: SELECT table1.column1, table1.column table2.column1,… FROM table1 RIGHT JOIN table2 ON table1.matching_column = table2.matching_column;



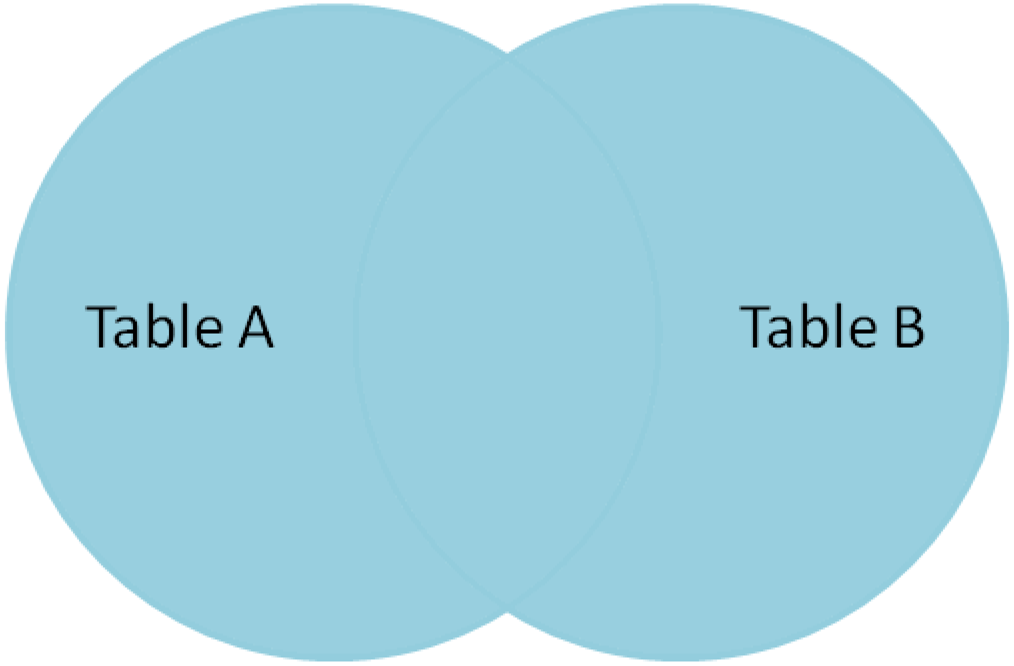
INNER JOIN

- The Inner Join keyword selects records that have matching values in both tables
- Syntax: SELECT table1.column1, table1.column2, table2.column1,… FROM table1 INNER JOIN table2 ON table1.matching_column = table2.matching_column;



FULL OUTER JOIN

- The Full Outer join returns all records when there is a match in either the left (1st) or the right (2nd) table
- For rows for which there is no matching, the result-set will contain *null* values
- Syntax: SELECT table1.column1, table1.column2, table2.column1,… FROM table1 FULL OUTER JOIN table2 ON table1.matching_column = table2.matching_column;



ACTIVITY

- Create the following tables and write a SQL statement to join the tables salesman, customer and orders so that the same column of each table appears once and only the relational rows are returned

Orders Table

Order_No	Purchase_amt	Order_date	Cust_ID	Salesman_ID
70001	150.5	2013-10-05	3005	5002
70009	270.65	2013-16-06	3001	5005
70014	65.26	2013-22-06	3009	5001
70019	110.5	2013-01-07	3015	5003
70020	948.81	2013-09-07	3017	5007

ACTIVITY

Customer Table

Customer_id	Cust_name	city	grade	Salesman_id
3002	Nick Fury	New York	100	5001
3007	Brad Davis	New York	200	5001
3005	Graham Zusi	Sacramento	200	5002
3008	Julian Green	Paris	300	5007

Salesman Table

Salesman_id	name	city	commission
5001	James Hoog	New York	0.15
5002	Nail Knite	Paris	0.13
5005	Pit Alex	London	0.11
5007	Paul Adam	Rome	0.13
5003	Randy Joe	Berlin	0.12